

Q1. Let $x_1, x_2, \dots, x_{10}$ be ten observations such that $\sum_{i=1}^{10} (x_i - 2) = 30$, $\sum_{i=1}^{10} (x_i - \beta)^2 = 98$, $\beta > 2$, and their variance is $\frac{4}{5}$. If μ and σ^2 are respectively the mean and the variance of $2(x_1 - 1) + 4\beta$,

$2(x_2 - 1) + 4\beta, \dots, 2(x_{10} - 1) + 4\beta$, then $\frac{\beta\mu}{\sigma^2}$ is equal to :

- (1) 100 (2) 120
(3) 110 (4) 90

Q2. Consider an A. P. of positive integers, whose sum of the first three terms is 54 and the sum of the first twenty terms lies between 1600 and 1800. Then its 11th term is :

- (1) 90 (2) 84
(3) 122 (4) 108

Q3. The number of solutions of the equation $\left(\frac{9}{x} - \frac{9}{\sqrt{x}} + 2\right)\left(\frac{2}{x} - \frac{7}{\sqrt{x}} + 3\right) = 0$ is:

- (1) 2 (2) 3
(3) 1 (4) 4

Q4. Define a relation R on the interval $[0, \frac{\pi}{2})$ by xRy if and only if $\sec^2 x - \tan^2 y = 1$. Then R is :

- (1) both reflexive and transitive but not symmetric (2) an equivalence relation
(3) reflexive but neither symmetric nor transitive (4) both reflexive and symmetric but not transitive

Q5. Two parabolas have the same focus (4, 3) and their directrices are the x-axis and the y-axis, respectively. If these parabolas intersect at the points A and B, then $(AB)^2$ is equal to :

- (1) 392 (2) 384
(3) 192 (4) 96

Q6. Let P be the set of seven digit numbers with sum of their digits equal to 11. If the numbers in P are formed by using the digits 1, 2 and 3 only, then the number of elements in the set P is :

- (1) 173 (2) 164
(3) 158 (4) 161

Q7. Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + 7\hat{j} + 3\hat{k}$. Let $L_1 : \vec{r} = (-\hat{i} + 2\hat{j} + \hat{k}) + \lambda\vec{a}, \lambda \in \mathbf{R}$ and $L_2 : \vec{r} = (\hat{j} + \hat{k}) + \mu\vec{b}, \mu \in \mathbf{R}$ be two lines. If the line L_3 passes through the point of intersection of L_1 and L_2 , and is parallel to $\vec{a} + \vec{b}$, then L_3 passes through the point :

- (1) (5, 17, 4) (2) (2, 8, 5)
(3) (8, 26, 12) (4) (-1, -1, 1)

Q8. Let $\vec{a} = 2\hat{i} - \hat{j} + 3\hat{k}$, $\vec{b} = 3\hat{i} - 5\hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} \times \vec{c} = \vec{c} \times \vec{b}$ and $(\vec{a} + \vec{c}) \cdot (\vec{b} + \vec{c}) = 168$. Then the maximum value of $|\vec{c}|^2$ is :

- (1) 462 (2) 77
(3) 154 (4) 308

Q9. The integral $80 \int_0^{\frac{\pi}{4}} \left(\frac{\sin \theta + \cos \theta}{9 + 16 \sin 2\theta} \right) d\theta$ is equal to :

- (1) $3 \log_e 4$ (2) $4 \log_e 3$
(3) $6 \log_e 4$ (4) $2 \log_e 3$

Q10. Let the ellipse $E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$ and $E_2 : \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$, $A < B$ have same eccentricity $\frac{1}{\sqrt{3}}$. Let the product of their lengths of latus rectums be $\frac{32}{\sqrt{3}}$, and the distance between the foci of E_1 be 4. If E_1 and E_2 meet at A, B, C and D , then the area of the quadrilateral $ABCD$ equals :

- (1) $\frac{12\sqrt{6}}{5}$ (2) $6\sqrt{6}$
 (3) $\frac{18\sqrt{6}}{5}$ (4) $\frac{24\sqrt{6}}{5}$

Q11. Let $A = [a_{ij}] = \begin{bmatrix} \log_5 128 & \log_4 5 \\ \log_5 8 & \log_4 25 \end{bmatrix}$. If A_{ij} is the cofactor of a_{ij} , $C_{ij} = \sum_{k=1}^2 a_{ik} A_{jk}$, $1 \leq i, j \leq 2$, and $C = [C_{ij}]$, then $8|C|$ is equal to :

- (1) 288 (2) 222
 (3) 242 (4) 262

Q12. Let $|z_1 - 8 - 2i| \leq 1$ and $|z_2 - 2 + 6i| \leq 2$, $z_1, z_2 \in \mathbf{C}$. Then the minimum value of $|z_1 - z_2|$ is :

- (1) 13 (2) 10
 (3) 3 (4) 7

Q13. Let $L_1 : \frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-1}{2}$ and $L_2 : \frac{x+1}{-1} = \frac{y-2}{2} = \frac{z}{1}$ be two lines. Let L_3 be a line passing through the point (α, β, γ) and be perpendicular to both L_1 and L_2 . If L_3 intersects L_1 , then $|5\alpha - 11\beta - 8\gamma|$ equals :

- (1) 20 (2) 18
 (3) 25 (4) 16

Q14. Let M and m respectively be the maximum and the minimum values of

$$f(x) = \begin{vmatrix} 1 + \sin^2 x & \cos^2 x & 4 \sin 4x \\ \sin^2 x & 1 + \cos^2 x & 4 \sin 4x \\ \sin^2 x & \cos^2 x & 1 + 4 \sin 4x \end{vmatrix}, x \in \mathbf{R} \text{ Then } M^4 - m^4 \text{ is equal to :}$$

- (1) 1280 (2) 1295
 (3) 1215 (4) 1040

Q15. Let ABC be a triangle formed by the lines $7x - 6y + 3 = 0$, $x + 2y - 31 = 0$ and $9x - 2y - 19 = 0$. Let the point (h, k) be the image of the centroid of $\triangle ABC$ in the line $3x + 6y - 53 = 0$. Then $h^2 + k^2 + hk$ is equal to:

- (1) 47 (2) 37
 (3) 36 (4) 40

Q16. The value of $\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{k^3 + 6k^2 + 11k + 5}{(k+3)!} \right)$ is:

- (1) $4/3$ (2) 2
 (3) $7/3$ (4) $5/3$

Q17. The least value of n for which the number of integral terms in the Binomial expansion of $(\sqrt[3]{7} + \sqrt[12]{11})^n$ is 183, is :

- (1) 2184 (2) 2196
 (3) 2148 (4) 2172